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Family and Parent-Child Relationship Correlates of Pediatric Cancer Survivors' Substance Use

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Abstract

Objective: The primary aims of this research were to examine substance use among adolescent and young adult survivors of pediatric cancer (AYA survivors) and AYA without a history of chronic or life threatening illness (AYA comparisons), and to explore links between demographic, medical, caregiver-AYA, and family-system factors with AYA substance use patterns.

Methods: Participants included 289 AYA (survivors, n=171; comparisons=118; 51% female; M_{age}=17.15, SD_{age}=2.86) and their caregivers (M_{age}=46.54, SD_{age}=6.81; 88% mothers). AYA and caregivers completed the Family Environment Scale and caregivers completed the Parenting Relationship Questionnaire at the initial assessment. Two years later, AYA completed an assessment of substance use. Chi-square and frequency analyses were used to compare differences in substance use among AYA survivors and comparisons. Multivariate analysis of variance was used to examine links between AYA substance use patterns with family and caregiver-AYA system level factors.

Results: Patterns of substance use did not differ between AYA survivors and comparisons. AYA survivors were more likely to report polysubstance use if caregivers endorsed problematic caregiver-AYA relationship patterns. Family functioning and caregiver relationship patterns did not predict AYA comparison substance use.

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Ethics approval. St. Jude Children's Research Hospital Institutional Review Board approved all study-related procedures.

Consent to participate. Prior to engaging in research activities, caregivers and adult participants provided written informed consent, and those who had not reached the age of majority provided verbal and/or written assent.

Consent for publication. Not Applicable

Data Availability Statement. The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

Code availability. Not Applicable

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Conclusion: AYA survivors were just as likely as AYA comparisons to engage in substance use, increasing their vulnerability to problematic health outcomes. Findings indicate the role caregiver-AYA relationship patterns may have on youth at-risk for substance use and potential mechanisms for future intervention.

Precis

AYA survivors of pediatric cancer are just as likely as healthy peers to engage in substance use behavior, increasing their vulnerability to problematic health outcomes. AYA-caregiver relationship functioning may be one modifiable target for substance use interventions within this population.

Keywords

AYA cancer survivor; pediatric cancer; childhood neoplasm; substance-related disorders; parent-child relations; family functioning

Adolescent and young adult pediatric cancer survivors (AYA survivors) are at elevated risk for developing secondary cancer diagnoses and other long-term health consequences as a result of their cancer treatment history [1, 2]. Though there are many immutable factors related to the increased vulnerability, health-promoting lifestyle behaviors, such as the avoidance of substance use, have been posited to reduce susceptibility to common late-effects [3], and as such, have been targeted in the Long-Term Follow-Up Guidelines outlined by the Children's Oncology Group [4].

Many risky health behaviors, such as drinking alcohol or smoking marijuana, are developmentally common manifestations of the experimentation and role exploration typical of the AYA period [5, 6]. Unfortunately, some of these behaviors can result in distinct and long-lasting physical harm among AYA survivors and/or within the general population [3, 7]. For example, marijuana and tobacco use can lead to pulmonary complications and cocaine use can lead to cardiovascular risks [3, 8]. For AYA survivors, the risk incurred by using substances likely only serves to exacerbate their existing vulnerability to long-term and late health effects due to their cancer history [3, 9, 10]. Nevertheless, many studies have shown that AYA survivors tend to engage in risky behaviors, like substance use, at rates similar to their healthy peers and siblings [5, 11], calling for the need to better elucidate factors linked to substance use for targeted intervention development.

Understanding risk and modifiable factors for substance use behaviors among AYA survivors has traditionally focused on individual-level characteristics including general mental health functioning [6], self-esteem [12], and depression and anxiety [13]. However, the AYA developmental literature points to the consideration of other social influences. In particular, the caregiver-AYA relationship and family system are important factors linked to AYA substance use [14–16]. In fact, in the general population, a recent meta-analytic review of 45 studies found family-based interventions to be more efficacious than interventions that targeted the AYA alone [17]. Specific factors that appear to be linked with AYA substance in the general population include caregiver-AYA relationship functioning domains such as

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attachment [18], communication [19], conflict [20], and involvement [21], as well as family functioning domains including conflict [22], cohesion [23], and expressiveness [24].

It is unclear, however, how prior research related to the family system and AYA substance use translates to AYA survivors. Prior research has documented family relationship patterns vary between families with and without a child with a chronic illness [25], thus warranting the need to better understand if and how the family social system may impact AYA survivor substance use. The primary aims of this research were twofold: first, to compare substance use behaviors among AYA survivors and AYA without a history of chronic or life threatening illness (AYA comparisons), and second, to explore longitudinal links between caregiver-AYA and family-system factors with subsequent substance use behavior, while controlling for demographic and medical factors. Consistent with prior research, we hypothesize that AYA survivors will exhibit similar levels of substance use as AYA comparisons. Second, AYA in both groups who have positive, high-quality relationships with their caregivers and family environments that are high in cohesion and expressiveness, and low in conflict will be less likely to use substances compared to AYA who have problematic caregiver relationships and family functioning patterns.

Methods

Procedures

Youth with cancer and their caregivers were approached during routine medical care visits as part of a larger, longitudinal study examining stress and adjustment. Data for this analysis were collected as part of the third (3 years from baseline) and fourth (5 years from baseline) time points of the larger study. At baseline, participants were recruited with consideration for diagnosis and based on one of four strata derived from time elapsed since their cancer diagnosis: 1–6 months; 6–24 months; 2–5 years; >5 years. Participants were eligible if they were (a) aged 8–17 years at the initial study visit, (b) could speak and read English, (c) did not have evidence of significant cognitive or sensory deficits, and (d) parent/legal guardian was willing to participate and provide consent for their child. A total of 378 children with cancer were approached regarding participation in the study, and 258 (68%) agreed to participate. Participants and non-participants did not differ by age, sex, or race/ethnicity, diagnostic category, or time since diagnosis.

Comparisons were recruited from local schools in a three-state area surrounding the hospital. Permission slips were distributed and included questions pertaining to child age, sex, race/ethnicity, and parental education and occupation. The returned data were used to create a pool of comparison participants, who were subsequently contacted, based on a demographic frequency matching procedure. Eligible participants were (a) 8–17 years of age; (b) able to speak and read English, (c) did not have any significant cognitive or sensory deficits, and (d) did not have a history of chronic or life-threatening illness. All procedures were approved by the hospital institutional review board. Of those in the pool who were contacted based on their match to a participating survivor, 86% consented to participate.

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Measures

Demographic and Clinical Variables.—The *Barratt Simplified Measure of Social Status* (BSMSS)[26] was used to create a socioeconomic status (SES) score based on caregivers' self-reported socio-demographic information. Participants were assigned a weighted score based on parents' level of educational attainment (scores ranging from 3–21) and current occupation (scores ranging from 5 to 45). Scores were summed and range between 8–66. For two-parent households, scores are averaged between caregivers. Higher scores indicate higher levels of SES. Participants' medical diagnosis, date of diagnosis, and treatment information were systematically collected through medical record review. The *Intensity of Treatment Rating Scale 2.0* (ITR-2)[27] provided an overall rating of participants' treatment intensity using diagnosis, stage/risk level, and treatment modality information. Demographic and clinical variables were collected and updated at each time point.

Substance Use.—A modified version of the *Practical Health Behavior Measure* [28], which was developed and derived from validated health-related questionnaires, was administered to AYA at time 4 (5 years post-baseline). For the present study, participants were asked to report how often during their lifetime they tried one or more of the following substances on a 5-point scale ranging from "Never" to "100 or more times": "Marijuana/Pot," "Cocaine/Crack," "Methamphetamine/Speed/Ice/Crystal Meth/Ecstasy," and "Prescription drugs that were not prescribed to you by a doctor." Given the positive skew distribution of all substance use items, use was categorized as "never used" or "used."

To evaluate risky drinking behaviors, participants reported the number of days in the past 30 days they engaged in binge drinking behaviors. Consistent with the Substance Abuse and Mental Health Services Administration, risky drinking was defined as ≥ 5 days of binge drinking (≥ 5 drinks consumed on occasion) within a 30 day range.

Cigarette smoking was categorized as "non-use,"" "former use," and "current use" consistent with procedures outlined in previous research [29]. As with other substance use categories, cigarette use was then categorized as "never used" and "used.".

Family Functioning.—The Family Environment Scale (FES)[30] was administered to caregivers and AYA at time 3. The original scale is a 90-item true-false inventory comprised of 10 subscales and 3 domains. Only the family relationship domain, which is comprised of Cohesion, Expressiveness, and Conflict subscales were collected, and items were converted to a 4-point Likert scale ranging from 1–4 (1=not at all like my family, 4=very much like my family). Internal reliabilities ranged from .63 to .78 in the present study.

Caregiver-AYA Relationship Quality.—Caregivers completed the *Parenting Relationship Questionnaire* (PRQ)[31] at time 3. This assessment consists of 71 items evaluating the caregiver-AYA relationship. The PRQ yields seven subscales, with the current study focused on the following: Attachment (feelings of closeness, empathy, and understanding), Communication (quality of information exchanged), Involvement (participation together in various activities), Parenting Confidence (comfort and confidence in parental role), and Relational Frustration (difficulties in relation to the identified child).

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With the exception of Relational Frustration, higher scores indicate better relationship quality [31]. The PRQ has demonstrated high internal consistency and moderate to high convergent validity [31] (α =.87 for the current study).

Planned Analyses

Substance use was collapsed into a single categorical variable, which included "non-use," "marijuana-only use," "cigarette-only use," " misuse-prescription-medicines-only," and "polysubstance use." No participant endorsed exclusive use of cocaine, crack, methamphetamine, speed, ice, crystal meth, ecstasy, or risky drinking; therefore, these categories were not created.

To determine frequency of use and compare the rate of substance use among AYA with cancer and comparisons, frequency and chi-square analyses were used. Associations among medical and demographic variables with AYA substance use categories were examined using chi-square and analysis of variance (ANOVA).

A total of four multivariate analysis of variance (MANOVA) were performed separately for survivors and comparisons and by AYA-caregiver relationship and family factors. Links between family functioning and caregiver-AYA relationship quality at the 3rd time point were examined with substance use categories evaluated at the 4th time point (2 years later), while controlling SES and age. For significant effects, follow-up analyses using one-way ANOVAs were performed. To account for multiple pairwise comparisons, the Bonferroni corrected p-value was used.

Results

Missing Data.

Of the 467 caregiver-AYA dyads enrolled at time 1, 8% of participants had missing data at time 3, 10% of participants had missing data at time 4, and 20% of participants had missing data at both time points, resulting in 289 (62%) participants with complete data at the time 3 and time 4 study visits. Reason for missing data included mortality, declining to participate, or loss to follow-up. Participants who had missing or incomplete data at either time point and participants with complete data (n=289) at both time points did not differ by sex, race, age of the caregiver, or SES measured at time 1. Participants with completed data at time 3 and time 4 were more likely to be older at time 1. For patient participants, treatment intensity rating, diagnostic category, and relapse status at time 1 did not differentiate the two groups. Participants who completed data at time 3 but not time 4 did not statistically differ across the aforementioned demographic categories. Secondary to the rate of missing data and links with participant age, data imputation was not conducted.

Participants.

Study participants included 171 AYA survivor-caregiver dyads and 118 AYA comparison-caregiver dyads. Demographic information can be found in Table 1. Differences did not emerge between AYA survivors and comparisons on age, race, sex, or caregiver age.

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Participants in the comparison dyads evidenced slightly higher time 4 SES scores than survivors.

Substance Use Patterns.

Patterns of substance use by demographic and medical variables are presented in Table 2, and patterns of substance use within the polysubstance use group are presented in Table 3. Patterns did not statistically differ between AYA survivors and AYA comparisons. Participant age statistically varied across substance use categories. Though the omnibus test for SES indicated differences across substance use categories, post-hoc analyses using the Bonferroni corrected p-value did not indicate statistically significant differences across groups (Table 2).

Associations between Caregiver-AYA Relationship Functioning and AYA Survivor Substance Use.

A one-way between groups MANOVA found a statistically significant difference on time 3 caregiver-AYA relationship functioning across time 4 substance use groups (F[20, 652]=1.72, p=.01; Pillai's Trace =.22; partial eta squared =.06) while controlling for SES and age.

When the results for the caregiver-AYA relationship functioning were examined separately, caregiver-report of involvement (F[4, 170]=2.75, p=.03; partial eta squared =.06), attachment (F[4, 170]=5.87, p<.001; partial eta squared =.13), communication (F[4,170]=5.22, p<.001; partial eta squared =.11), confidence (F[4, 170]=3.04, p=.02; partial eta squared =.07), and relational frustration (F[4, 170]=3.35, p=.01; partial eta squared =.08) were statistically significant.

To further probe differences in AYA-caregiver relationship quality across substance use categories, follow-up one-way ANOVAs were performed (see Table 4). In brief, AYA survivors who reported polysubstance use had caregivers who reported lower levels of caregiver-AYA attachment as compared to every other substance use category. Further, caregiver report of communication within this group was lower than every other substance use category with the exception of AYA survivors who reported misuse-prescription-medicines-only. Finally, caregiver report of relational frustration was higher and caregiver involvement and confidence were lower among AYA survivors who reported polysubstance use as compared to those who reported non-use.

Associations between Family Functioning and AYA Survivor Substance Use.

A one-way between groups MANOVA did not find statistically significant difference on time 3 family functioning variables across time 4 substance use groups among AYA survivors (F[24,596]=1.49, p=.06; Pillai's Trace =.23; partial eta squared =.06).

Associations between Caregiver-AYA Relationship Functioning and AYA Comparison Substance Use.

A one-way between groups MANOVA did not find a statistically significant difference on time 3 caregiver-AYA relationship functioning across time 4 substance use groups in the

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AYA comparison group (F[20, 440]=1.36, p=.14; Pillai's Trace =.23; partial eta squared =.06).

Associations between Family Functioning and AYA Comparison Substance Use.

A one-way between groups MANOVA did not find statistically significant differences on time 3 family functioning variables across time 4 substance use groups in the AYA comparison group (F [24, 380] = .18, p=.81; Pillai's Trace =.18; partial eta squared = .05).

Discussion

Risky substance use behaviors are developmentally common among AYA [5, 6]. Similar to prior research and consistent with our hypothesis, findings from this research indicate AYA survivors are equally vulnerable to engagement in substance use as their peers [5, 6, 11]. In general, five patterns of substance use emerged for AYA survivors and comparisons: non-use, marijuana-only use, cigarette-only use, misuse-prescription-medicines-only, and polysubstance use, with a substantial minority engaging in some type of substance use. Given the increased health-related complications secondary to substance use in an already at-risk population, our additional aims were to illuminate potential risk and modifiable factors among AYA survivors and examine how these might differ from a cohort of AYA comparisons for future development of targeted interventions.

Demographic and treatment-related risk factors explored were not linked to AYA substance use patterns across both groups, with the exception of age. Not surprisingly, older AYA in both groups were more likely to report marijuana-only use or polysubstance use, as compared to those who engaged in non-use or misuse-prescription-medicines-only groups. The use of non-prescribed medication continues to be a growing problem among younger adolescents [32], with recent research indicating younger AYA were more susceptible to misuse of prescription medication than older AYA [33]. As this area of research evolves, it will be imperative to understand the types of non-prescribed medications AYA survivors are likely to misuse as well as how this group is gaining access to these medications. This information will be imperative for the development of screening tools and interventions to mitigate problematic use.

Polysubstance use was the most endorsed substance use category among both AYA survivors and comparisons. Prior research has documented among the general population that polysubstance use can increase susceptibility to problematic health and social achievement outcomes [34, 35]. This is particularly troublesome for AYA survivors who are already at increased risk for detrimental outcomes within these domains [3, 9, 36]. Findings from the present study suggest targeting the caregiver-AYA relationship may be one modifiable mechanism to reduce polysubstance use behavior. Indeed, prior research has documented therapeutic interventions that repair attachment with the caregiver, increase positive communication between AYA and the caregiver, foster caregiver confidence, support age-appropriate caregiver involvement, and defuse relational frustration may decrease substance use among AYA engaging in polysubstance use [17–19, 21, 37].

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It is interesting that AYA survivors who reported non-use, cigarette-only use, marijuana-only use, and misuse-prescription-medicines-only use did not differ on any AYA-caregiver relationship patterns. This finding may indicate ruptured caregiver-AYA relationship patterns may be more robust predictors of polysubstance use, and factors not evaluated in the present study, such as peer and parental substance use are more indicative of other substance use patterns [38]. Further research is needed to clarify these findings

Several findings emerged that were inconsistent with our hypotheses. For both groups, family functioning did not differ across substance use categories. Further, caregiver-AYA relationship quality was not linked to comparison's substance use behavior. In the broader literature, the influence of family functioning has been primarily found in studies focused on treatment-seeking families [17]. This may indicate a selection bias in the population targeted for this study, or perhaps a larger base rate of substance use behaviors is needed to appreciate the impact of family functioning on these behaviors.

Limitations and Future Directions

Findings from this research should be considered in light of several limitations. First, all substance-use data were self-reported, and may reflect underreporting of actual substance use behaviors. As this research continues to grow, considerations for objective data of substance use will be important for gaining a true reflection of substance use patterns. Relatedly, only caregivers reported on relationship functioning, and of those caregivers, most were mothers. Receiving reports from multiple informants may best assist in understanding correlates of AYA substance use. Second, substance use and polysubstance use were loosely defined, as the length and timing of overlap of substance use were not assessed and categorizations of use (i.e., "use," "non-use") limited the variability of substance use behavior that was examined. As a result, participants who experimented with substances once were potentially grouped with individuals who used a given substance more frequently. Precisely defining substance use will be critical in the development of this area of research. Third, this research only targeted the family system, and considerations for individual-level factors (e.g., mental health) and other social systems (e.g., peers) were not included. During the AYA development period, mental health status [6, 12], peer relations, [39]and other social influences [38] become increasingly influential on AYA substance use behaviors. As such, examining the multiplicative impact of individual-level factors with other social systems will be a fruitful avenue for further investigation. Fourth, it is difficult to establish temporal sequencing between caregiver-AYA functioning and substance use. The design of this study assumed a unidirectional relationship; however, it is plausible substance use behaviors impact caregiver-AYA and family functioning behaviors or that this relationship may be reciprocal in nature. Fifth, inclusion and exclusion criteria for this study (e.g., English-speaking, lack of cognitive deficit) as well as loss to follow-up, may have systematically biased results. Finally, while this study includes a sample that falls within the AYA age range, the AYA within this study do not fall in line with the National Cancer Institute's (NCI) broadly defined AYA category (15–39 years of age), and findings likely do not generalize to the upper reaches of this population. However, the NCI age range is very broad and social systems, particularly familial, in older AYA likely contribute to substance use in qualitatively different ways than in younger AYA. It will be imperative that similar

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research be conducted with older AYA with consideration made for their developmental stage.

Clinical Implications

Nearly one third of AYA survivors in this study endorsed substance use, with the vast majority reporting polysubstance use behaviors. This finding in conjunction with AYA survivorship literature [40] supports the need to routinely screen for substance use patterns and provide appropriate education and referrals as needed. Further, considering polysubstance use was the most common category of substance use, screening and subsequent interventions should target multiple substances versus a narrow focus on one substance. Though findings from this research should be considered preliminary, evidence from the general AYA population indicate referral for therapeutic interventions that target caregiver-AYA relationship may provide an effective avenue to reduce polysubstance use behaviors.

Conclusion

A substantial minority of AYA survivors are engaging in substance use, with patterns mirroring that of their same-aged peers, increasing their vulnerability to future health related complications. Findings from this research indicate the potential role caregiver-AYA relationship patterns have on youth at risk for multiple substance use and provide a possible mechanism for future intervention.

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Table 1.

Demographic and medical factors across groups at Time 4

	AYA survivors *n* = 171	AYA Comparison *n* = 118
Sex		
% Female	50	53
% Male	50	47
Time 4 Age		
Mean (SD)	17.30 (2.91)	16.93 (2.80)
Range	12–23	13–23
Race		
% White	76	77
% Black	19	20
% Other	5	3
Parent Type		
% Mom	85	91
% Dad	11	9
% Other	4	0
Caregiver Age		
Mean (SD)	46.65 (6.86)	46.56 (6.79)
Range	27–72	31–68
Time 4 Socioeconomic Status Score		
Mean (SD)	42.78 (11.86)	46.53 (11.46)
Range	11–66	12–66
Diagnosis		
% Leukemia	33	--
% Hodgkin's & Non-Hodgkin's Lymphoma	13	--
% Solid Tumor	39	--
% Brain Tumor	15	--
% Healthy Comparison	--	100.0
Years Since Diagnosis at Time 4		
Mean (SD)	8.92 (4.20)	--
Treatment Intensity		
% Least Intensive	7	--
% Moderately Intensive	39	--
% Very Intensive	26	--
% Most Intensive	27	--

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Table 2.

Demographic and Medical Factors Across Substance Use Categories

	Non-Use (1)		Cigarette Only Use (2)		Marijuana-Only Use (3)		Misuse- Prescription-Medicine-Only (4)		Polysubstance Use (5)		Inferential Test	p-value	Post-Hoc Comparison*
	M/%	SD	M/%	SD	M/%	SD	M/%	SD	M/%	SD			
Group											$\chi^2(4, N=289) =5.24$	.26	
% AYA Survivors	69		7		6		4		15				
% AYA Comparisons	66		3		9		8		14				
Sex											$\chi^2(4, N=289) =5.34$	.25	-
% Male	63		8		6		6		16				
% Female	72		3		8		4		12				
Race											$\chi^2(8, N=289) =5.03$	.76	-
% White	67		6		7		5		15				
% Black	70		5		11		5		9				
% Other	67		0		0		8		25				
Socioeconomic Status	45.27	10.97	40.19	14.40	47.04	10.32	13.07	13.57	39.71	13.67	$F(4, 288) =2.74$	.03	-
Participant Age	16.37	2.70	18.13	2.63	19.29	1.82	16.07	2.49	19.80	1.95	$F(4, 288) =20.85$	<.001	1< 3, 5; 4 < 3,5
Diagnosis											$\chi^2(12, N=171) =17.59$	.13	
% Leukemia	73		2		9		0		16				
% Hodgkin's & Non-Hodgkin's Lymphoma	64		9		9		5		14				
% Solid Tumor	60		10		5		8		18				
% Brain Tumor	89		8		0		0		4				
Years Since Diagnosis at Time 4	8.91	4.13	9.56	5.60	8.33	3.50	6.74	1.67	9.45	4.50	$F(4, 170) =.62$	.65	-
Treatment Intensity											$\chi^2(12, N=163) =19.54$	.08	
% Least Intensive	55		18		0		0		27				
% Moderately Intensive	66		5		11		2		17				
% Very Intensive	65		7		5		12		12				

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	Non-Use (1)		Cigarette Only Use (2)		Marijuana-Only Use (3)		Misuse- Prescription-Medicine-Only (4)		Polysubstance Use (5)		Inferential Test	p-value	Post-Hoc Comparison *
	M/%	SD	M/%	SD	M/%	SD	M/%	SD	M/%	SD			
% Most Intensive	76		9		2		0		13				

Note. Percentages rounded to the nearest whole number. AYA= Adolescent and Young Adult,

* = significant differences in at least $p<.05$ using the Bonferroni correction.

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Table 3.

Substance use patterns among AYA who engage in polysubstance use

	Cigarette Use (n=41)	Marijuana/Pot Use (n=41)	Misuse-Prescription-Medicine-Only (n=20)	Methamphetamine Use (n=7)	Cocaine/Crack Use (n=7)	Risky Drinking (n=4)
Current Cigarette Use	-	90% (n=37)	80% (n=16)	100% (n=7)	100% (n=7)	100% (n=4)
Marijuana/Pot Use	90% (n=37)	-	100% (n=20)	100% (n=7)	100% (n=7)	100% (n=4)
Misuse-Prescription-Medicine-Only	39% (n=16)	49% (n=20)	-	100% (n=7)	86% (n=6)	50% (n=2)
Methamphetamine Use	17% (n=7)	17% (n=7)	35% (n=7)	-	64% (n=5)	25% (n=1)
Cocaine/Crack Use	17% (n=7)	17% (n=7)	30% (n=6)	71% (n=5)	-	25% (n=1)
Risky Drinking	10% (n=4)	10% (n=4)	10% (n=2)	14% (n=1)	14% (n=1)	-

Note. Percentages calculated based on the number of participants endorsing use within each category. Columns will not add up to 100 given overlapping use within each category.

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Table 4.

Post-hoc analyses examining mean differences in caregiver-AYA functioning across substance use categories among AYA survivors.

	Non-Use (1)		Cigarette-Only Use (2)		Marijuana-Only Use (3)		Misuse- Prescription-Medicine-Only (4)		Polysubstance Use (5)		Post-Hoc Comparison*
	M	SD	M	SD	M	SD	M	SD	M	SD	
Caregiver-AYA Involvement	11.69	4.25	11.92	5.33	10.30	2.71	12.00	6.20	7.52	4.92	1>5
Caregiver-AYA Attachment	21.97	4.11	22.33	3.73	23.90	3.96	24.67	5.24	17.64	5.28	1,2,3,4>5
Caregiver-AYA Communication	20.34	4.29	21.17	3.46	21.20	2.35	20.33	2.25	15.68	5.74	1,2,3>5
Caregiver Parenting Confidence	16.88	3.12	17.08	2.87	16.80	2.62	17.50	3.27	14.16	4.48	1>5
Caregiver-AYA Relational Frustration	7.81	4.54	6.83	4.61	7.60	3.50	7.00	3.03	10.60	4.46	1<5

Note. AYA= Adolescent and Young Adult,

*= significant differences in at least $p<.01$.